EPSR-test-ex108 Pairs missing from the .pcof; ncoeffp given in the .EPSR.inp

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises how the `epsr` engine [1,2,3] completes an incomplete potential file: atom pairs absent from the `.pcof` get default coefficients, and the number of EP coefficients to read is taken from the `.EPSR.inp` when the `.pcof` has no `ncoeffp` line. This is what happens when a user edits or adds atom types by hand.

Code exercised

Fortran (original, fortran_src/):

- `read_and_save_files.f`: `read_pcof` (new pair sets, `npairspcof` 20 and 21; reading only `ncoeffp` coefficients), `save_files_others`
- `process_symbols.f`: `check_symbol` (`ncoeffp` block)
- `epsr_standalone_rev3.f`: `inita` (resets `ncoeffp` from `rminpt`/`rmaxpt`)

Rust (port, rust/epsr-core/src/):

- `pcof.rs`: `read_pcof`, `write_pcof_pairs`
- `symbols.rs`: `check_symbol`; `structure.rs`: `inita`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å^3). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. For this case the DMSO S - water H1 and water O1-H1 pairs are missing: they start with a zero empirical potential (`pcoefp` = `(1, 0, ...)`) and `rminex` = `max(rminfac*dcor, roverlap)` (0 here).

Inputs

`scripts/make_inputs.py`: S-H1, O1-H1 and the `ncoeffp` header line removed from the `.pcof` (19 pairs left); `ncoeffp 60` inserted in the `.EPSR.inp` before `fnameato`; `ntimes 2`, `niter 2`. `commands.txt`: `epsr 200DMSO400Water.EPSR.inp`, `repeat 2`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
Only 60 coefficients per pair are read in pass 1 (61-116 stay 0); `inita` then sets `ncoeffp` to 116 and the stored Fortran reference `.pcof` has `ncoeffp 116` and 21 pairs, with S-H1 and O1-H1 appended as pairs 20 and 21 with `coulfac 0.00000`. The README reports gcov evidence: `npairspcof=npairspcof+1` executed twice, the `check_symbol` `ncoeffp` block once.

Notes

Fortran quirk reproduced: for a new pair `coulfac`, `coordex` and `efacp` are not set (COMMON, 0), so the pair has no Coulomb term and its EP is not refined in the run that creates it; it is refined from pass 2 on.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.